

OpenClaw AI Weather Station Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], and [05-Three Dialogue Configuration Tests] in the `Preparation Before Use` directory. If you are only using TUI/WhatsApp, you can skip `04`. This article assumes the basic environment is ready and only covers the AI weather station project itself.

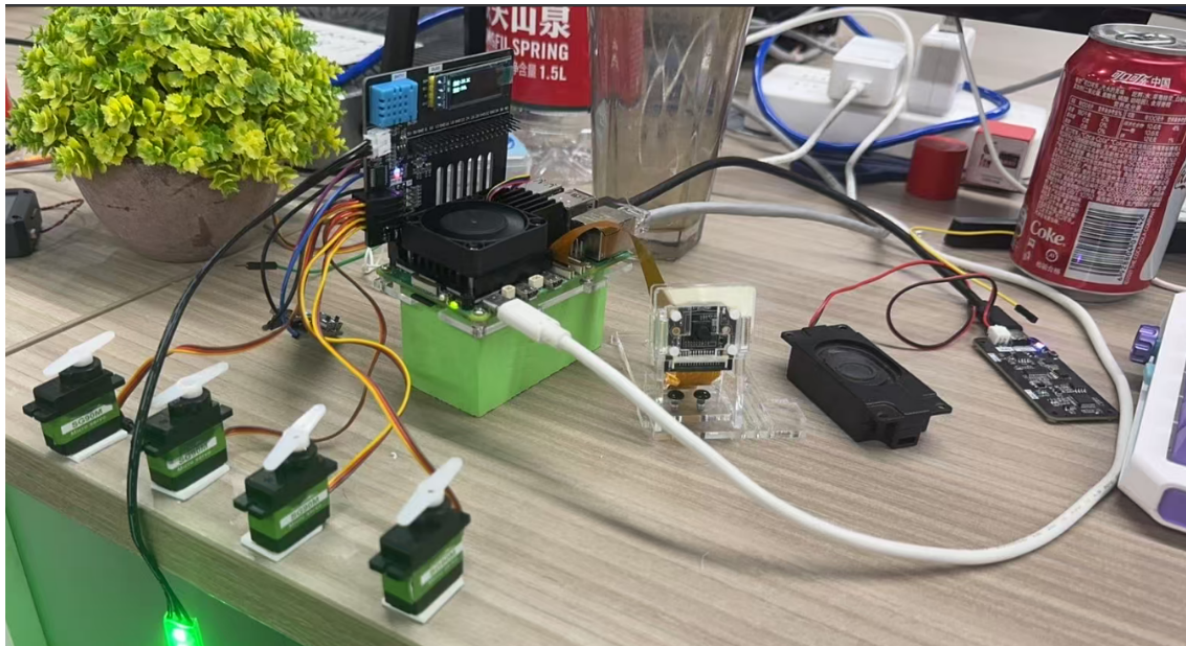
1. Tutorial Objectives

This article demonstrates OpenClaw's capability in the desktop weather station scenario through temperature and humidity collection, information display, and AI explanation. After completing this tutorial, readers will be able to:

- Understand how to combine DHT, OLED, and RGB into a local AI weather station
- Communicate with OpenClaw via WhatsApp, TUI, and voice commands
- Complete four project scenarios: environmental collection, AI broadcast, weather card, status warning

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display screen
- RGB light strip / RGB LED module
- [Optional] AI voice module + speaker
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial:

- Temperature and humidity reading command: `/home/pi/hardware_ha1_demo/dht`
- OLED control command: `/home/pi/hardware_ha1_demo/oled`

- RGB control command: `/home/pi/hardware_hal_demo/rgb`
- Voice entry: `/home/pi/voice_to_openclaw/src/main.py`
- Terminal text entry: `/home/pi/voice_to_openclaw/src/text_main.py`

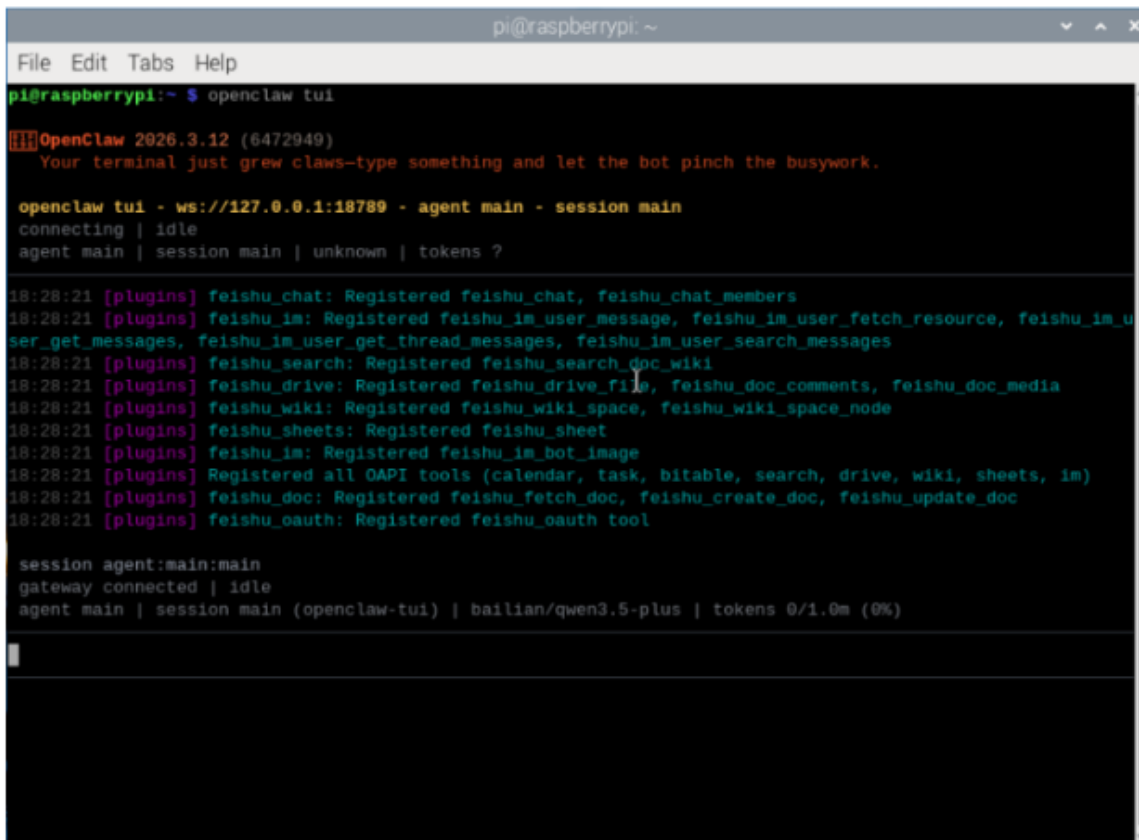
The focus of this tutorial is "local environment weather station", which is based on DHT indoor temperature and humidity data for display and explanation;

4. Testing Project Basic Capabilities

Before starting this project, please complete an `openclaw tui` basic conversation self-check first; if you haven't done this yet, see [05-Three Dialogue Configuration Tests]. After passing, proceed with the AI weather station-specific test.

It is recommended to first enter OpenClaw TUI:

```
openclaw tui
```



```
pi@raspberrypi: ~
File Edit Tabs Help
pi@raspberrypi:~ $ openclaw tui
[!!!]OpenClaw 2026.3.12 (6472949)
  Your terminal just grew claws-type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im_bot_image: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)
```

Then test the following types of commands in sequence:

- Read current temperature and humidity
- Display weather card on OLED
- If environment is normal, light up green light
- Give me a brief environmental broadcast

5. Three Ways to Communicate with OpenClaw

Method	Suitable Scenario	Advantages	Suggested Use
WhatsApp	Remote environment viewing, classroom demonstration	Suitable for remotely asking "how is the environment now"	Demonstrate "remote weather station assistant"
TUI	Local debugging, threshold verification	Easiest to troubleshoot sensor and display logic	First to get the collection chain working
Voice Code	Voice broadcast, desktop assistant	More like daily usage experience	As the final demonstration solution

If you have the optional AI voice module + speaker, the TUI and WhatsApp conversations need to broadcast OpenClaw's replies. You can run the following program in the terminal. This is not needed for AI voice module conversations.

- Modify the TTS enable parameter

```
/home/pi/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast replies. After saving, run the program below in the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) pi@raspberrypi:~/voice_to_openclaw $ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/pi/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [05-Three Dialogue Configuration Tests]. After creating, you can remotely check the environment status in WhatsApp, for example:

- How is the indoor environment now? If it's too hot, remind me

20:34



< 2



+86 153 3885 7526 (你)

给自己发消息

...waiting for me...

20:33 ✓✓

How is the indoor environment now? If it is too hot, remind me.

20:34 ✓✓

Current indoor environment readings:



Temperature: 29.7°C (85.5°F)



Humidity: 54%

Assessment: The temperature is **just below the hot threshold** (30°C). It's warm but not excessively hot yet.

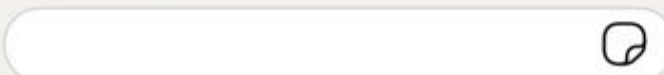
Recommendation:

- The room is comfortably warm, but if the temperature rises above 30°C, I'll remind you with a red light warning
- Current humidity at 54% is in a comfortable range (40-60% is ideal)
- Consider ventilation if you feel warm or if temperature continues to rise

Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically.

20:34 ✓✓



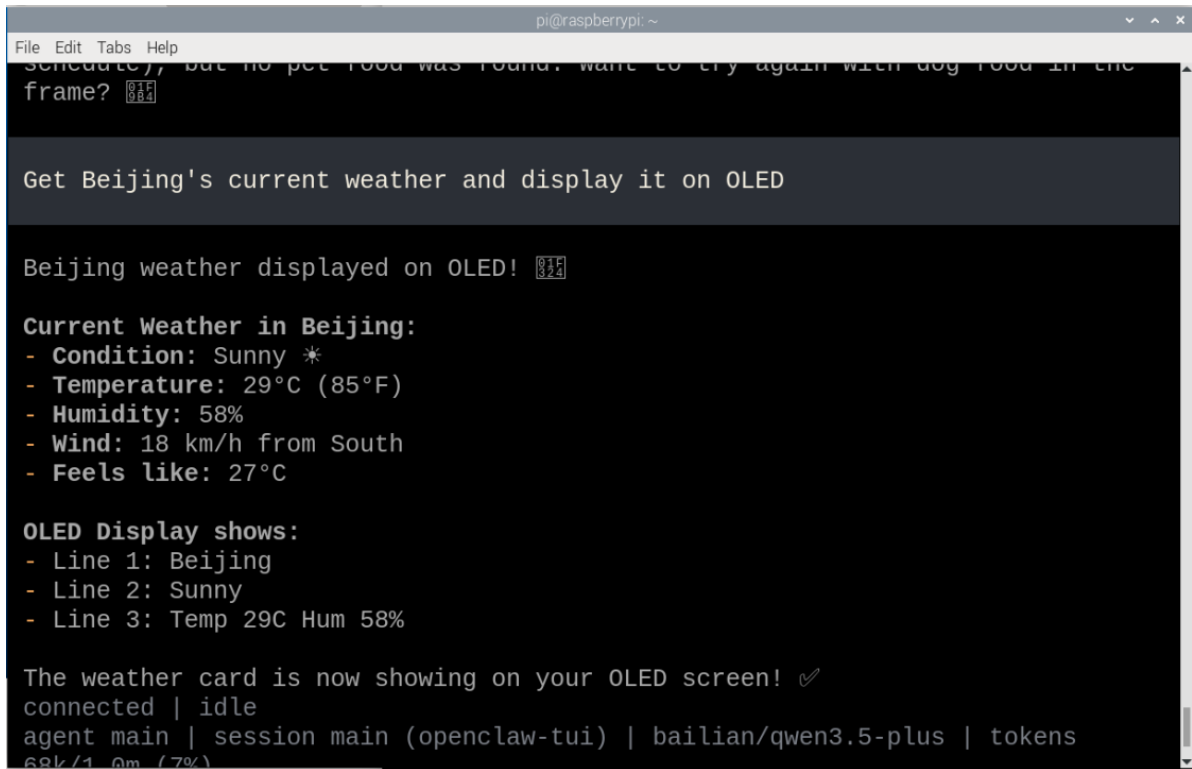
5.2 TUI Solution

If you have already completed the TUI self-check from chapter 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then input project-related commands, for example:

- Get Beijing's current weather and display it on OLED



```
pi@raspberrypi: ~  
File Edit Tabs Help  
Schedule, but no pet food was found. Want to try again with dog food in the  
frame? [0.4] [0.4]  
  
Get Beijing's current weather and display it on OLED  
  
Beijing weather displayed on OLED! [0.4] [0.4]  
  
Current Weather in Beijing:  
- Condition: Sunny *  
- Temperature: 29°C (85°F)  
- Humidity: 58%  
- Wind: 18 km/h from South  
- Feels like: 27°C  
  
OLED Display shows:  
- Line 1: Beijing  
- Line 2: Sunny  
- Line 3: Temp 29C Hum 58%  
  
The weather card is now showing on your OLED screen! ✓  
connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens  
68k/1.0m (7%)
```

5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up flow, please refer directly to [04-AI Voice Interaction Configuration]. After waking up, you can directly speak control content, for example:

- AI weather station


```
pi@raspberrypi: ~/voice_to_openclaw
File Edit Tabs Help
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号，开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音，正在转写

[你] Ai weather station.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] ## 🌤️ AI Weather Station Complete!

**Local Sensor Readings:**
-
-      **Indoor Temperature:** 30°C (86°F)
- **Indoor Humidity:** 51%
-      **Outdoor Weather (Shenzhen):**
- **Condition:** Partly cloudy 🌤️
-
-      **Outdoor Temperature:** 28°C (82°F)
- **Feels Like:** 35°C (95°F)
-
-      **Outdoor Humidity:** 84%
- **Wind:** 9 km/h from South
- **Rain
-      Chance:** 72% today

**3-Day Forecast:**
- **Today (May 14):** High
-      28°C, Low 23°C - Patchy rain nearby
- **Tomorrow (May 15):** High
-      26°C, Low 23°C - Moderate rain expected
```

6. Project Implementation

Note: All demonstrations below use the `openclaw tui` method for conversation. You can also choose WhatsApp or voice to complete.

6.1 Scenario 1: Environmental Collection

Experiment Objective

Read the temperature and humidity of the local environment to complete the most basic data sampling for the weather station.

Effect Description

OpenClaw will call DHT to get current ambient data and return the results to the user, forming a real-time "weather reading" of the indoor environment.

Example Commands

- Read current ambient temperature and humidity
- What are the current indoor temperature and humidity
- Help me check current weather data

6.2 Scenario 2: AI Broadcast

Experiment Objective

Have OpenClaw, not just report numbers, but also summarize the environment status like a weather broadcaster.

Effect Description

OpenClaw can summarize "26 degrees, 57% humidity" into more natural phrases such as "environment is quite comfortable", "a bit muggy", "air is a bit dry", etc.

Example Commands

- Give me a brief environmental broadcast
- Is the current environment comfortable
- Make a weather summary based on current readings

6.3 Scenario 3: OLED Weather Card

Experiment Objective

Display the local temperature and humidity results as a page more like a "weather card".

Effect Description

Besides pure text results, OLED can also arrange location, status, temperature, and humidity into a concise card, making the entire project more like a displayable desktop device.

Example Commands

- Display an indoor weather card on OLED
- Get current weather for xx location and display it on OLED

6.4 Scenario 4: RGB Status Warning

Experiment Objective

Quickly express the current environment status with RGB colors.

Effect Description

For example, light up green when environment is normal, yellow when it's too hot, and red when abnormal. This way, even without looking at text, you can quickly know what state the current environment is in.

Example Commands

- Light up green light if environment is normal
- Light up yellow light if temperature is high
- Light up red light and flash if abnormal